

Advanced SQL Concepts

Constraints

Normalization

Locks

Triggers

SQL Jobs

Constraints

- Rules that ensure valid data.
- • PRIMARY KEY
- • FOREIGN KEY
- • NOT NULL
- • UNIQUE
- • CHECK
- • DEFAULT

Constraints Example

- CREATE TABLE Students (
 - StudentID INT PRIMARY KEY,
 - Name VARCHAR(100) NOT NULL,
 - Age INT CHECK (Age>=18)
-);

Normalization

- Organizes data to reduce redundancy.
- 1NF: Atomic values
- 2NF: Remove partial dependency
- 3NF: Remove transitive dependency

Locks

- Protect data during concurrent access.
- • Shared Lock
- • Exclusive Lock
- • Update Lock
- • Intent Lock

Triggers

- Automatically execute after INSERT, UPDATE, or DELETE.
- Example:
- CREATE TRIGGER trg_Insert
- ON Employees
- AFTER INSERT
- AS PRINT 'Inserted';

SQL Jobs

- Automate scheduled tasks.
- • Database Backup
- • Index Maintenance
- • Report Generation
- • Data Import

Summary

- Constraints → Data Integrity
- Normalization → Less Redundancy
- Locks → Concurrency Control
- Triggers → Automation
- Jobs → Scheduling

Thank You

- Questions?